

DICT-AFOT

ADVANCED FIBRE OPTIC TRAINER WITH AM, FM, PPM & PWM



FEATURES

- Circuit diagram printed on PCB
- On-board Sine Wave Generator
- On-board Carrier Signal Generator
- On-board AM, FM, PPM & PWM
- Provision for voice communication through Microphone & Speaker/Headphone
- PC to PC communication
- On-board Fault Switches
- Test point provided to study the signal at different stages

TECHNICAL SPECIFICATIONS

- Fiber optic LED's having peak wavelength of Emission 660nm
- Inbuilt Signal Source : Analog & Digital
Bandwidth : 350KHz
Digital Bandwidth : 2.5MHz
Filters Transmitter : 02 nos.
- Receiver : 02 Nos. Fiber Optic Photo Detector.
- Modulation Techniques : Amplitude Modulation, Frequency Modulation & Pulse width Modulation
- Drivers : 2 nos. with Analog & Digital modes.
- Analog : 02 nos. 4th order Butterworth Filter.
- Cut Off Frequency 3.4 KHz
- Onboard Function Generator : 1KHz variable amplitude sine wave 1KHz TL square wave
- Voice Line : Fiber optic voice link using microphone & speaker (built-in).
- PC to PC communication : Using 2 channel through RS-232 serial port.
- Port : 19200 baud.
- Type of cable : Step indexed multimode PMMA Plastic cable
Core Refractive index : 1.492
Clad Refractive index : 1.406
NA : Better than 0.5
Acpt. Angle : Better than 60°
Fiber Dia : 1000mu.
Outer Dia. : 2.2mm
Length : 1m, 0.5m
- PLL Detector : 1 nos.
- Comparators : 02 nos.
- Switch Faults : Provided on board.
- Test Points : 50
- Interconnection : 2/4mm banana sockets
- DC Supply : Built in IC regulated power supplies
- 220V $\pm 10\%$, 50Hz mains operated
- Enclosed in an attractive ABS plastic cabinet with cover
- Standard Accessories : User's Manual with patch cords

DICT-AFT

ANALOG FIBRE OPTIC TRAINER



TECHNICAL SPECIFICATIONS

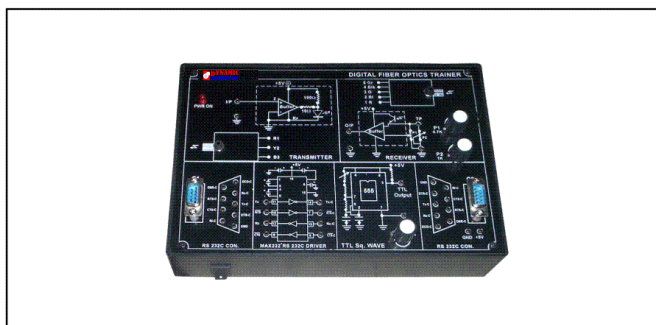
- Fibre Optic Analog Transmitter and Receiver
- 600nm or 850nm SMA Connectorised FO LED
- Optical to Electrical & Electrical to Optical Converter
- SMA Connectorised FO Photo Transistor
- 1Mtr. PMMA (960/1000 u) SMA FO Cable
- Experiment for Attenuation in fibre optic cable
- Transmission and Reception of audio signals using Microphone
- Interconnection : 2/4mm banana sockets
- DC Supply : Built in IC regulated power supplies
- 220V $\pm 10\%$, 50Hz mains operated
- Enclosed in an attractive ABS plastic cabinet with cover
- Standard Accessories : User's Manual with patch cords

Optional Accessories :

- Audio Amplifier with MIC

DICT-DFT

DIGITAL FIBRE OPTIC TRAINER



TECHNICAL SPECIFICATIONS

- Fibre Optic Digital Transmitter/Receiver Module
- 600nm or 850nm SMA Connectorised FO LED
- RS 232C Driver using MAX-232-2 Nos
- Optical to Electrical & Electrical to Optical Converter
- SMA Connectorised FO Photo Transistor
- 1Mtr. PMMA SMA FO Cable
- RS232C cables for communication with PC (Optional).
- Interconnection : 2/4mm banana sockets
- DC Supply : Built in IC regulated power supplies
- 220V $\pm 10\%$, 50Hz mains operated
- Enclosed in an attractive ABS plastic cabinet with cover
- Standard Accessories : User's Manual with patch cords

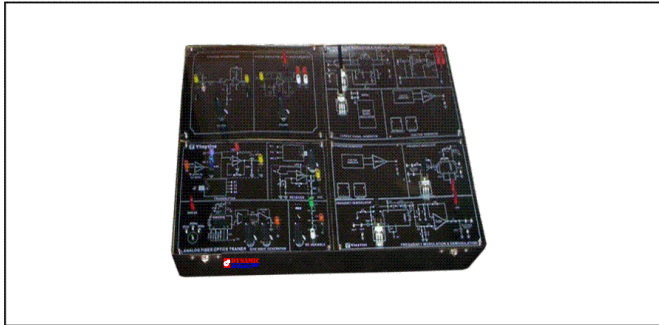
Specifications are subject to change without notice due to our constant efforts for improvement.

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DICT-FOCT

FIBRE OPTIC COMMUNICATION TRAINER



FEATURES

- Circuit diagram printed on PCB.
- Transformer based power supply.
- Strongly support by systematic operating instructions.
- Enclosed in an attractive wooden box with cover

TECHNICAL SPECIFICATIONS

- Onboard Sine Wave Generation with variable amplitude 0-5Vp-p and frequency 20Hz to 20KHz
- Onboard Variable DC supply from -5V to +5V
- Onboard Carrier Generation with variable amplitude 0-10Vp-p and frequency 60KHz (Fixed)
- Study of Fiber Optic Communication Trainer through :

Analog Fiber Optic :

- Separate blocks for Transmitter & Receiver

Amplitude Modulation & Demodulation :

- Separate blocks for Modulation using IC 1496 & Demodulation using IC TL084
- Onboard potentiometer for varying depth/percent of modulation

Frequency Modulation & Demodulation Section :

- Separate blocks for Modulation using IC TL084 & 8038 and Demodulation using IC LM565 & TL084

Dynamic Microphone & Audio Amplifier with Speaker Section :

- Separate blocks for dynamic microphone & Audio Amplifier with loud speaker
- Separate potentiometer provided for controlling the intensity of sound level in both section

General Specifications :

- On-board inbuilt 2mm socket for observing the signal and waveform at different stages
- Onboard LED for Power indication.
- 220 V, Built in IC regulated power supply
- Enclosed in attractive ABS plastic cabinet with cover to protect onboard components
- User's manual and patch chords

DICT-FOT

FIBRE OPTIC TRAINER WITH POWER METER



FEATURES

- Circuit diagram printed on PCB.
- Transformer based power supply
- Onboard Power Meter
- Provision for Numerical Aperture
- Onboard RS-232C Driver
- Strongly support by systematic operating instructions.
- Enclosed in an attractive wooden box with cover

TECHNICAL SPECIFICATIONS

- Onboard Sine Wave Generation with variable amplitude 0-5Vp-p and frequency 10Hz to 200KHz
- Onboard Power Meter with facility of input for Fibre optic cable
- Onboard Variable DC supply from -5V to +5V
- Onboard Provision for setup for numeric aperture (optional)
- RS 232C Driver using MAX-232-2 Nos
- RS232C cables for communication with PC (Optional)

Analog Fiber Optic :

- Separate blocks for Transmitter & Receiver
- Transmission and Reception of audio signals using Microphone

Digital Fiber Optic :

- Separate blocks for Transmitter & Receiver

General Specifications :

- On-board inbuilt 2mm socket for observing the signal and waveform at different stages
- Onboard LED for Power indication.
- 220 V, Built in IC regulated power supply
- Enclosed in attractive ABS plastic cabinet with cover to protect onboard components
- User's manual and patch chords

DIAL-APA

AUDIO AMPLIFIER WITH MIC

SPECIFICATIONS

- Circuit diagram printed on PCB to ease of understand the circuit
- Onboard Dynamic Microphone with MIC
- Onboard Audio Amplifier with Loud Speaker
- Facility to control volume through potentiometer
- DC Supply : Built in IC regulated power supplies
- 220V $\pm 10\%$, 50Hz mains operated
- Enclosed in an attractive ABS plastic cabinet with cover

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DICL-DBM

OPTICAL POWER METER

SPECIFICATIONS

- Detector : Silicon detector
- Range : 0 dBm to 60 dBm
- Display : 12mm LCD
- Wavelength : 660 & 950nm*
- Connector : SMA
- Accuracy : 0.5 dBm
- Power : 9V Battery

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